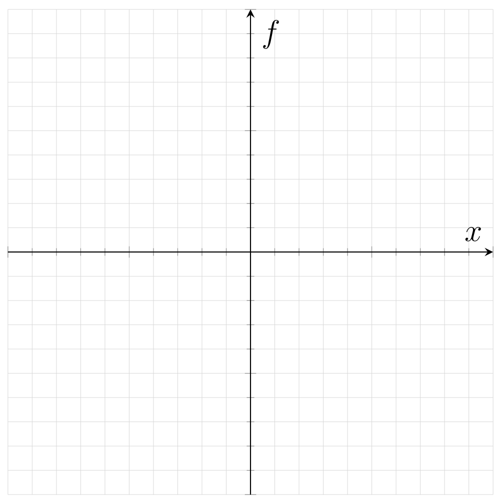
1 Quotient/reciprocal functions

Functions where there is another function in the denominator eg:

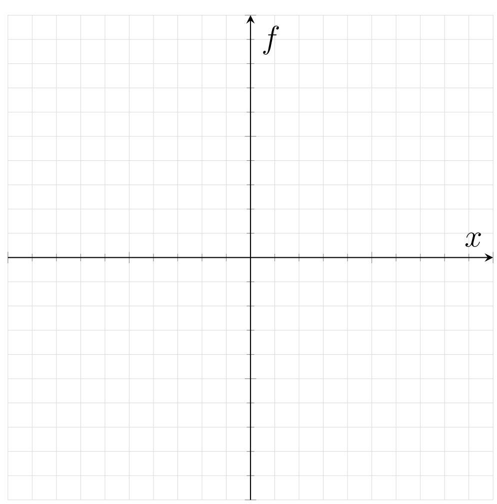
$$f(x) = \frac{1}{2x^2 - 4x + 3}$$

Try plotting it:



Try plotting this by hand:

$$y = \frac{1}{x^2 - 4x + 4}$$



Strategy?

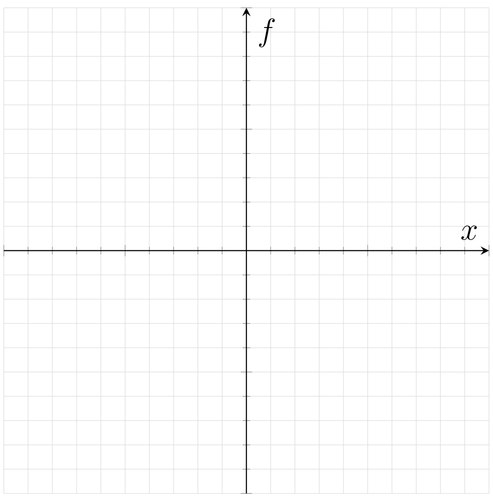
- Plot denominator first
- Gauge whether the function shoots up to $+\infty$ or $-\infty$ based on whether the denominator is positive or negative

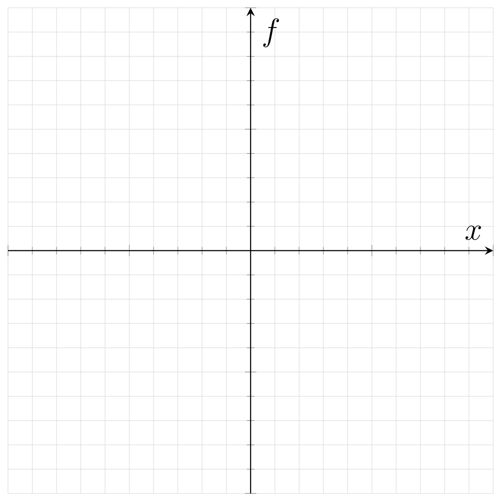
1.1 Notable reciprocal functions

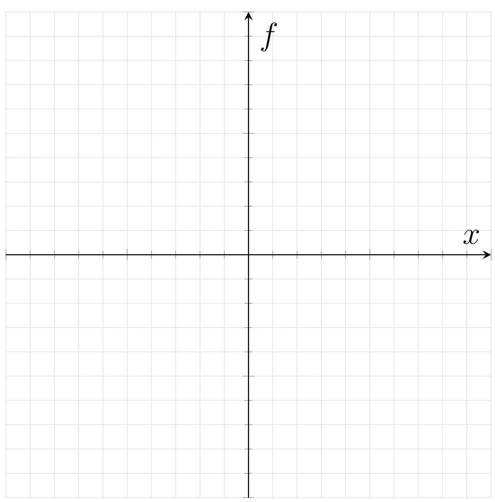
Trigonometric reciprocal functions: (Plot each one, note where the asymptotes are)

- $\frac{1}{\sin(x)} = \csc(x)$
- $\frac{1}{\cos(x)} = \sec(x)$
- $\frac{1}{\tan(x)} = \cot(x)$

(give them new names so as to not mistake them for inverses)





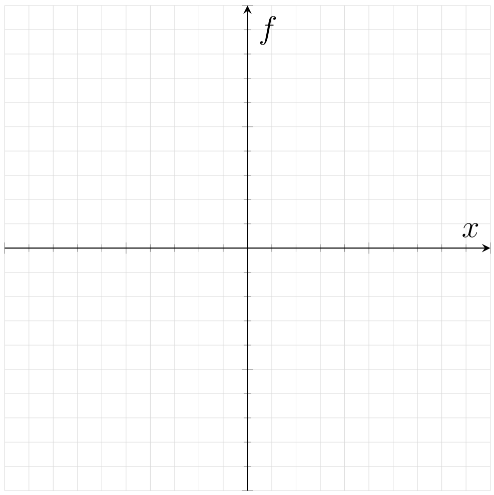


Transformations work the same as normal functions

1.2 Transformations on reciprocal trig functions

Plot the following over $(-\pi, 2\pi)$

- $f(x) = 2\csc(x+3)$



- $g(x) = -\frac{1}{2}\sec(2x - \frac{\pi}{4}) + 1$